

Consolidated Priority Matrices v2: Cybersecurity & Cyber-Resilience Strategy for the Sovereign Cloud–Edge–IoT Continuum

Iraklis Symeonidis

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Abstract

This document presents the consolidated priority matrices for the European Union’s cybersecurity and cyber-resilience strategy across the Cloud–Edge–IoT continuum. The matrices organize ten strategic priorities across three clusters (Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty) and track their progression through three phases (2025–2030, 2030–2035, 2035–2040). Each priority is assessed across four analytical dimensions: (1) R&D Maturity, (2) Policy/Regulatory/Standards, (3) Operational Deployment & Adoption, and (4) Market Uptake & Strategic Sovereignty.

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1 Introduction

This document contains three priority progression matrices structured to align with the generalized methodology framework. Each matrix analyzes strategic priorities across three phases and four analytical dimensions, with critical enabler (gate condition) rows indicating the necessary conditions for phase transitions.

2 Priority Cluster 1: Sovereign Trusted Infrastructure

Table 1: Progression Matrix — Priority Cluster 1: Sovereign Trusted Infrastructure (P1: Hardware RoT, P2: PQC Agility, P3: Lightweight Crypto & PhySec, P8: Confidential Computing, P9: Island Mode). Each phase is assessed across four analytical dimensions. Critical Enabler rows state the single named gate condition controlling the transition into the following phase.

dimHdr	Priority Cluster 1 — Sovereign Trusted Infrastructure — Sub-priorities: P1, P2, P3, P8, P9
dimHdr Analytical Dimension	Phase 1: Foundation & Compliance (2025–2030)
dimRI (1) R&I Maturity	RISC-V secure element prototypes validated against EUCC evaluation criteria under EU Chips Act pilot funding; NIST FIPS 203/204/205 algorithms optimised for 8/16-bit microcontrollers within 32 KB RAM constraints; RF fingerprinting CSI/CFO device authentication validated in controlled mMTC environments; RISC-V Keystone TEE attestation protocols tested against heterogeneous edge fleet configurations; soft-state Island Mode credential caching validated in 5G disconnection simulation environments. The cluster ODP is Applied R&D — the lowest common denominator driven by P2, P3, and P8; P1 and P9 have reached Fragmented Piloting and are not undeployed. <i>Transition criterion: at least one functional prototype per sub-priority validated against the minimum EU compliance baseline (NIS2, CRA, or AI Act as applicable).</i>

dimHdr Analytical Dimension	Phase 1 (2025–2030)
dimPol (2) Policy, Regulatory & Standards	CRA Articles 10 and 11 full enforcement from October 2027 establishes the mandatory market condition for hardware security certification investment. NIS2 Article 21(2)(e) cryptographic policy obligations active from October 2024. NIST FIPS 203/204/205 finalized August 2024 providing algorithm standard basis; ENISA PQC transition guidelines in publication; EUCC evaluation methodology available but no EU-manufactured secure element has completed High assurance evaluation; 3GPP TS 33.501 Release 18 provides 5G security architecture for Island Mode integration. DORA Article 9 ICT resilience testing requirements apply to P9; AI Act Article 9 risk management obligations apply to TEE-attested autonomous workloads. <i>Critical policy gap: no EU-specific ratified standard for RISC-V secure element security evaluation existed at the time of writing; this must be addressed in Phase 1 CEN/CENELEC JTC 13 work programme.</i>

dimHdr Analytical Dimension	Phase 1 (2025–2030)
dimOps (3) Operational Deployment & Adoption	Intel TDX and AMD SEV-SNP dominate all production TEE deployments in EU cloud infrastructure; RISC-V secure elements at prototype stage in 3–5 EU-funded research consortia; edge nodes universally operate with hard cloud dependencies and no validated autonomous fallback in production; hybrid PQC pilots operate in cloud research environments but not in IoT field deployments; Island Mode soft-state protocols validated in 5G research corridors. Deployment actors are limited to large institutional early adopters (national cybersecurity agencies, telecom operators in EU corridor pilots, EU-funded research consortia); SME adoption is absent due to tooling immaturity. <i>Transition criterion: documented operational deployment across at least two EU critical infrastructure sectors or Member State jurisdictions with published findings.</i>

dimHdr Analytical Dimension	Phase 1 (2025–2030)
dimMkt (4) Market Uptake & Strategic Sovereignty SSL: High Foreign Dependency	Non-EU TPMs hold >90% of EU critical infrastructure secure element market; EU Chips Act fabrication pilot lines initiated but first certified secure elements are 3–5 years from production; no EU-manufactured EUCC ‘High’ certified component available in any product category; PQC implementation ecosystem dominated by US/UK cryptography research outputs; Island Mode tooling absent from EU commercial product portfolios. <i>SSL baseline: High Foreign Dependency across all five sub-priorities. Market transition criterion: at least one EU-designed solution achieves a verifiable market entry event (product launch, procurement award, or EUCC/EUCS certification).</i>
dimGate <i>Critical Enabler — Phase 1 → Phase 2</i>	dimGate CRA Articles 10 and 11 enforcement (October 2027) creating mandatory market demand for hardware security certification and establishing the commercial conditions that justify EU Chips Act fabrication investment and EUCC ‘High’ evaluation resource allocation.
dimHdr	<i>Sub-Priority Decomposition</i> — individual maturity tracking per constituent priority across phases.
dimSub <i>P1: Hardware Roots of Trust & EUCC ‘High’</i>	ODP: Fragmented Piloting SSL: High Foreign Dependency
dimSub <i>P2: PQC Agility for Edge/IoT</i>	ODP: Applied R&D SSL: High Foreign Dependency
dimSub <i>P3: Lightweight Crypto & Physical-Layer Security</i>	ODP: Applied R&D SSL: High Foreign Dependency
dimSub <i>P8: Confidential Computing & Privacy-Preserving AI</i>	ODP: Applied R&D SSL: High Foreign Dependency

dimHdr Analytical Dimension	Phase 1 (2025–2030)
dimSub <i>P9: Resilient-by-Design & Autonomous Island Mode</i>	ODP: Fragmented Piloting SSL: High Foreign Dependency

Appendix: Cluster Matrix Notes

The priority matrices presented above organize strategic priorities across three phases (2025–2030, 2030–2035, 2035–2040) and four analytical dimensions: R&D Maturity, Policy/Regulatory/Standards, Operational Deployment & Adoption, and Market Uptake & Strategic Sovereignty.

Critical Enabler rows identify the gate conditions controlling phase transitions. Each condition must be met to justify investment and resource allocation in the following phase.